

Unit 2, Lesson 6: Dividing Multi-Digit Numbers with Decimals to the Thousandths

Benchmark

MA.6.NSO.2.1 Multiply and divide positive multi-digit numbers with decimals to the thousandths, including using a standard algorithm with procedural fluency.

Learning Target

- ☐ I can fluently divide multi-digit numbers with decimals to the thousandths.

2.6.1 Warm-Up: Quotients as Decimals

The quotients of several division expressions are shown.

$1 \div 2 = 0.5$	$1 \div 3 = 0.333\overline{3}$	$1 \div 4 = 0.25$
$1 \div 5 = 0.2$	$1 \div 6 = 0.666\overline{6}$	$1 \div 7 = 0.14285\overline{7}$
$1 \div 8 = 0.125$	$1 \div 9 = 0.111\overline{1}$	$1 \div 10 = 0.1$
$1 \div 11 = 0.0909\overline{09}$	$1 \div 12 = 0.083\overline{3}$	$1 \div 13 = 0.07692\overline{3}$

1. Write something you notice about the values of the quotients and something you still wonder.

I notice...	I wonder...

2.6.2 Refresher: Long Division

Jiang was dividing using long division and got stuck.

1. Complete the problem to show Jiang the missing steps.

$$\begin{array}{r}
 15 \overline{) 1875} \\
 \underline{- 15} \\
 37 \\
 \underline{- 35} \\
 20 \\
 \underline{- 20} \\
 0
 \end{array}$$

2. What advice about long division do you have that could help Jiang in the future?

2.6.3 Exploration: Dividing Decimals

Giovanny and Alina used different strategies to find the quotient of $48.78 \div 9$.

Giovanny’s Strategy

$$\begin{array}{r} 5.42 \\ 9 \overline{) 48.78} \\ \underline{- 45} \\ 37 \\ \underline{- 36} \\ 18 \\ \underline{- 18} \\ 0 \end{array}$$

Alina’s Strategy

$$\begin{array}{r} 5.42 \\ 900 \overline{) 4878} \\ \underline{- 4500} \\ 3780 \\ \underline{- 3600} \\ 1800 \\ \underline{- 1800} \\ 00 \end{array}$$

$48.78 \div 9 = 5.42$

Work with your partner to answer the questions about Giovanny and Alina’s strategies.

1. How are the two students’ calculations alike and different?

Similarities	Differences

2. Look at Giovanni's calculations. Complete the statements with the appropriate place value to explain how you can tell that the 36 means "36 tenths" and the 18 means "18 hundredths" in his calculations.
- The 3 in 36 is in the _____ place and the 6 is in the _____ place, so 36 in this calculation means 36 tenths.
 - The 1 in 18 is in the _____ place, so it has the same value as 10 hundredths. The 8 is in the _____ place. Together, they make 18 hundredths.
3. Look at Alina's calculations. Complete the statement to describe the values of 3600 and 1800 in her calculation.
- In Alina's calculations, the 3600 represents 3600 _____, and the 1800 represents 1800 _____.
4. Why is the value of $48.78 \div 9$ the same as the value of $4878 \div 900$?
5. Explain why $51.84 \div 6.4$ has the same value as $518.4 \div 64$.
6. Find the value of $51.84 \div 6.4$ using long division. Show your work.

2.6.4 Bring It Together: Dividing Decimals

1. Complete the summary of the exploration of dividing decimals using the words and phrases from the bank below.

does not change	multiply
power of 10	whole numbers

One way to find a quotient of two decimals is to first

_____ each decimal by a _____

so that both values are _____. If we

multiply both values by the same power of 10, this

_____ **the value of the quotient.**

2.6.5 Guided Practice: Dividing Decimals

1. Find each quotient using a strategy similar to Giovanni's or Alina's. Show your work.
 - a. $24.2 \div 1.1$
 - b. $17.028 \div 0.08$
2. Coby and Olivia each found the quotient of $10.92 \div 2.6$. The values they each used are listed.
 - Coby calculated $1092 \div 260$.
 - Olivia calculated $109.2 \div 26$.
 - a. Will the values they used result in the same quotient as $10.92 \div 2.6$? Explain.
 - b. Find the quotient of $10.92 \div 2.6$.

2.6.6 Your Turn!

1. Find each quotient. Show your work.

a. $43.396 \div 57.1$

b. $29.88 \div 0.36$

c. $262.5 \div 5.25$

d. $15.129 \div 12.3$

2.6.7 Wrap-Up: Reflection

1. Draw an **X** on the line to indicate where your current learning is in relation to the learning target.

I can fluently divide multi-digit numbers with decimals to the thousandths.

I need help.
I can't get started.

I'm getting there.

I understand
and have evidence.



2. Briefly explain why you placed the X where you did.

